

Audit Quality, Corporate Governance, and Earnings Management: A Meta-Analysis

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Earnings management is of great concern to corporate stakeholders. While numerous studies have investigated the effects of various corporate governance and audit quality variables on earnings management, empirical evidence is rather inconsistent. This meta-analysis identifies 12 significant relationships by integrating results from 48 prior studies. For corporate governance, the independence of the board of directors and its expertise have a negative relationship with earnings management. Similar negative relationships exist between earnings management and the audit committee's independence, its size, expertise, and the number of meetings. The audit committee's share ownership has a positive effect on earnings management. For audit quality, auditor tenure, auditor size, and specialization have a negative relationship with earnings management. Auditor independence, as measured by fee ratio and total fee, is also a deterrent to earnings management.

Key words: Audit committee, audit quality, auditor choice, corporate governance, earnings management, fraud, independence, meta-analysis

SUMMARY

Earnings management is of great concern to corporate stakeholders. While numerous studies have investigated the effects of various corporate governance and audit quality variables on earnings management, empirical evidence of their effects is rather inconsistent. This paper applied meta-analytic techniques to empirical data from 48 studies that examined relationships between

corporate governance and audit quality variables and earnings management. Of the 17 relationships tested, 12 showed significant effects. Specifically, for corporate governance, the independence of the board of directors and its expertise have a negative relationship with earnings management. Similar negative relationships exist between earnings management and the audit committee's independence, its size, expertise, and the number of meetings. The audit committee's share ownership is positively related to earnings management. For audit quality, auditor tenure, auditor size, and auditor specialization have a

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negative relationship with earnings management. Auditor independence, as measured by fee ratio and total fee, is also a deterrent to earnings management.

Relationships that were non-significant include the effects of the board of directors' stock ownership, existence of an audit committee, and the separation of the board chairperson position from the CEO position. These are potential areas for future research. Another prospective research stream is to examine the moderating effect of certain variables such as country. For example, we found that while the independence of the board of directors is significant overall, the effect is more profound in countries other than the United States. The opposite is observed about the independence of the audit committee: the effect is more pronounced in the United States than in other countries. Similarly, the effect of auditor size as a deterrent to earnings management is more significant in the United States than in other countries. In fact, the effect is not significant when data from other countries are tested. On the other hand, the share ownership of the board of directors has no significant effect on earnings management, in either the United States or other countries. We also tested the effects of two levels of audit committee independence (complete and proportional independence), and found both to have a significant effect in reducing earnings management. Similarly, future research can examine the moderating effect of important regulations, such as the Sarbanes-Oxley Act, by comparing data from pre- and post-SOX.

INTRODUCTION

Much research has been conducted on the determinants of earnings management such as a firm's financial characteristics, corporate governance and audit quality. However, the extant studies have reported mixed results. The purpose of this paper is to use meta-analysis techniques to synthesize and evaluate the findings from the large number of existing studies on the determinants of earnings management. Our focus is on the effect of corporate governance effectiveness and audit quality. Meta-analysis is the application of statistical methods to a large collection of results from existing individual studies for the purpose of integrating and evaluating the research findings. Use of meta-analysis often makes it possible to reach stronger conclusions or more valid inferences

about a common research issue than in a narrative literary review (Wolf, 1986). The remainder of this paper is organized as follows. The next section provides an overview of prior research on the relationships between earnings management and corporate governance and audit quality. Next, we describe the research methodology, followed by discussions of meta-analytic results. The last section presents concluding remarks.

LITERATURE REVIEW

Earnings management

Various definitions exist for earnings management. Schipper (1989) appears to have captured the essence of earnings management by defining it as 'purposeful intervention in the external financial reporting process with the intent of obtaining private gain'. Likewise, Healy & Wahlen (1999) state that 'earnings management occurs when managers use judgment in financial reporting and in structuring transactions to alter financial reports to either mislead some stakeholders about the underlying economic performance of the company or to influence contractual outcomes that depend on reported accounting numbers.' Regardless of the definition adopted, earnings management is inherently unobservable. Most prior studies use various measures of discretionary or abnormal accruals as proxies for earnings management. Other measures used include earnings restatement and financial reporting fraud. A regression model is typically employed to investigate the effects of various independent variables on earnings management in the form of:

$$EM_k = \beta_0 + \beta_1 X_{1,k} + \beta_2 X_{2,k} + \dots + \beta_i X_{i,k} + \varepsilon_k, \quad (1)$$

$$k = 1, 2, \dots, N$$

where EM is earnings management and X_i represents either a control variable or an independent variable under investigation in study k in a set of N prior studies in a meta-analysis. Next, we provide an overview of research on the effects on earnings management of factors relating to corporate governance effectiveness and quality of external audit.

Corporate governance

Owing to the separation of ownership and control (and the resulting agency problems) in the modern business world, a system of corporate governance

is necessary, through which management is overseen and supervised to reduce the agency costs and align the interests of management with those of the investors. While there is no generally accepted definition, corporate governance may be defined as a system 'consisting of all the people, processes and activities to help ensure stewardship over an entity's assets' (Messier *et al.*, 2008: 36). A good corporate governance structure helps ensure that the management properly utilizes the enterprise's resources in the best interest of absentee owners, and fairly reports the financial condition and operating performance of the enterprise. For corporations in the US, the body primarily responsible for management oversight is the board of directors and its designated committees. The audit committee, consisting of members of the board, assists the board in its oversight of the financial reporting process.

The role of the corporate governance structure in financial reporting is to ensure compliance with generally accepted accounting principles (GAAP) and to maintain the credibility of corporate financial statements. The corporate governance mechanisms that are the focus of recent regulations and prior studies are attributes related to the organization and functioning of the board in general and its audit committee in particular. Properly structured corporate governance mechanisms are expected to reduce earnings management because they provide effective monitoring of management in the financial reporting process.

Unfortunately, empirical research to date provides inconsistent evidence on the relationship between measures of corporate governance effectiveness and earnings management (earnings quality or the lack thereof). For example, while Davidson *et al.* (2005) and Klein (2002) report a significantly negative relationship between board independence and earnings management, Park & Shin (2004) and Peasnell *et al.* (2005) fail to find any significant relationship. Such inconsistency also exists in empirical evidence on the relationships between earnings management and other attributes related to board effectiveness in monitoring management in the financial reporting process.

Often the board of directors delegates work on important tasks to its standing committees. For example, the audit committee is charged with overseeing financial reporting. The audit committee's primary role is to help ensure high

quality financial reporting by the firm. Therefore, a properly structured and functioning audit committee is expected to reduce opportunistic earnings management. A number of recent studies examine the effect of an audit committee's characteristics on earnings management but have provided mixed evidence as is the case in research on effectiveness of the board of directors in reducing earnings management. For example, while Abbott *et al.* (2000) document that occurrence of earnings management decreases with independence of the audit committee, Choi *et al.* (2004) find no such effect. Also, Xie *et al.* (2003) find no significant association between the number of directors on the audit committee and earnings management. Similarly, Abbott *et al.* (2004) find no impact of audit committee size on earnings restatements. In contrast, Yang & Krishnan (2005) report that audit committee size is negatively associated with earnings management (using abnormal accrual as proxy), implying that a certain minimum number of audit committee members may be relevant to quality of financial reporting.

There is also concern that compensating audit committee directors with stock and stock options may result in impairment of their independence (Millstein, 2002); however, empirical evidence on this issue has been limited until recently. Bédard *et al.* (2004) document that the more stock options that can be exercised in the short run relative to the total of options and stocks held by audit committee directors, the higher the likelihood of aggressive earnings management. Yang & Krishnan (2005) report that stock ownership by board members on the audit committee is positively associated with earnings management. These results contradict the findings by Beasley (1996) that the likelihood of fraud decreases as stock ownership by outside directors (not necessarily audit committee directors) on the board increases.

Audit quality

The agency problems associated with the separation of ownership and control, along with information asymmetry between management and absentee owners, create the demand for external audit. External auditors are responsible for verifying that the financial statements are fairly stated in conformity with GAAP and that these statements reflect the 'true' economic condition and operating results of the entity. Thus, the external auditor's verification adds credibility to

the company's financial statements. Also, the external auditors are required by auditing standards to discuss and communicate with the audit committee about the quality, not just the acceptability, of accounting principles applied by the client company. Therefore, a quality audit is expected to constrain opportunistic earnings management as well as to reduce information risk that the financial reports contain material misstatements or omissions.

The guidelines and measures for the quality of the external auditor's performance are set forth in generally accepted auditing standards, such as competence, independence and exercise of due professional care. Obviously, the quality of the auditor's performance is multi-dimensional as set forth in the auditing standards, and differences in audit quality are to be expected. 'Audit quality differences result in variation in credibility offered by the auditors, and in the earnings quality of their audit clients. Because auditor quality is multidimensional and inherently unobservable, no single auditor characteristic can be used to proxy for it' (Balsam *et al.*, 2003: 71). Since audit quality may be affected by a number of factors, it is not surprising that researchers have used various measures to proxy for audit quality in prior studies.

For example, researchers have examined the effects of auditor brand name (auditor size) and industry specialization, auditor tenure, provision of various services by the auditor and auditor independence on a number of issues directly or indirectly related to financial reporting. Empirical evidence on these audit quality measures has been mixed. For example, while many existing studies show that the use of brand name (i.e., Big 4/5/6) auditors reduces earnings management (e.g., Becker *et al.*, 1998; Francis *et al.*, 1999; Lin *et al.*, 2006), many others fail to report such findings (e.g., Bédard *et al.*, 2004; Davidson *et al.*, 2005). As another example, Frankel *et al.* (2002) report that the ratio of non-audit service fees to total auditors' fees (proxy for impaired auditor independence) is positively associated with small earnings surprises and with the magnitude of discretionary accruals (proxies for earnings quality or earnings management). Their results provide support to the SEC's position that non-audit fees can impair auditor independence and hence audit quality. On the other hand, Chung & Kallapur (2003) find no significant relationship between discretionary accruals and audit fees or non-audit fees. Similarly, Raghunandan *et al.* (2003) find no evidence

supporting the claim that non-audit fees or total fees inappropriately influence the audit of financial statements that are subsequently restated. Inconsistent results reported in prior studies about the effects of the other factors affecting audit quality on earnings quality are highlighted in the results section below.

METHODOLOGY

The first step in a meta-analysis is to locate relevant studies through computer and manual searches. Various combinations of key words are used to search commonly available computerized literature databases, such as ABI/Inform and Business Source Premier, to locate empirical (archival) studies that deal with earnings management. Key words used include earnings, accrual, restatement, fraud, management, quality, audit and governance. The computer searches were conducted from late October to early November 2007 and publications available online or in print were then reviewed for possible inclusion. References in studies identified in computer searches were also scanned to find additional studies. Published articles had to be empirical archival studies that met all of the following criteria for inclusion in the meta-analysis: (1) the dependent variable must be measures based on abnormal (discretionary) accrual, financial statement restatement or reporting fraud; (2) the independent variables in the empirical multivariate model must include measures of audit quality, or effectiveness of corporate governance relating to the board of directors or audit committee; and (3) the test statistics or *p*-value needed to compute the effect size must be reported.

Ultimately, 48 studies were included in the meta-analysis. Table 1 lists these studies and also provides information about the dependent and independent variables, and data years and countries for sample firms used. Many of these included studies measure the same variable in multiple ways. For example, audit committee independence can be measured by its membership that is made of 100 percent outsiders (i.e., completely independent) or over 50 percent outsiders (i.e., proportionally independent) (Klein, 2002). Multiple results from the same study are combined to satisfy the independent sample requirement for meta-analysis. We also examine sample size, data years and countries used in the included studies to ensure no later studies use identical data from any of the prior publications. A

Table 1: Studies included

<i>Study</i>	<i>Independent variable</i>	<i>Dependent variable</i>	<i>Data country/year</i>
Abbott <i>et al.</i> 2000	audit committee characteristics	fraud	US 1980–96
Abbott <i>et al.</i> 2004	audit committee characteristics	restatement	US 1991–99
Agrawal & Chadha 2005	auditor fees; board & audit committee characteristics	restatement	US 2000–01
Antle <i>et al.</i> 2006	auditor fees	abnormal accrual	UK 1994–2000
Ashbaugh <i>et al.</i> 2003	non-audit service	abnormal accrual	US 2001
Balsam <i>et al.</i> 2003	auditor specialization	abnormal accrual	US 1991–99
Bauwhede <i>et al.</i> 2003	auditor size	discretionary accrual	Belgium 1991–97
Bauwhede & Willekens 2004	auditor size	discretionary accrual	Belgium 1994–96
Beasley 1996	audit committee characteristics	fraud	US 1979–90
Becker <i>et al.</i> 1998	audit committee and BOD characteristics	abnormal accrual	US 1993
Bédard <i>et al.</i> 2004	audit committee characteristics	abnormal accrual	US 1996
Benkel <i>et al.</i> 2006	BOD & audit committee characteristics	discretionary accrual	Australia 2001–03
Carey & Simnett 2006	audit partner tenure	abnormal accrual	Australia 1995
Chen <i>et al.</i> 2005	auditor size & specialist	discretionary accrual	Taiwan 1999–2002
Choi <i>et al.</i> 2004	audit committee characteristics	abnormal accrual	Korea 2000–01
Chung <i>et al.</i> 2005	audit committee & BOD characteristics	abnormal accrual	US 1980–96
Cormier & Martinez 2006	BOD & auditor size	discretionary accrual	France 2000–02
Crutchley <i>et al.</i> 2007	BOD characteristics	fraud	US 1991–2002
Davidson <i>et al.</i> 2005	audit committee characteristics	abnormal accrual	Australia 2000
Ferguson <i>et al.</i> 2004	non-audit service	abnormal accrual	UK 1996–98
Firth <i>et al.</i> 2007	board characteristics & auditor size	discretionary accrual	China 1998–2003
Francis <i>et al.</i> 1999	auditor size	discretionary accrual	US 1975–94
Frankel <i>et al.</i> 2002	non-audit fee	abnormal accrual	US 2001
Gul <i>et al.</i> 2002	audit committee characteristics	abnormal accrual	Australia 1992–93
Hoitash <i>et al.</i> 2007	auditor fees & size	discretionary accrual	US 2000–03
Huang <i>et al.</i> 2007	auditor fees & size	discretionary accrual	US 2003–04
Jaggi & Leung 2007	BOD characteristics & auditor size	discretionary accrual	Hong Kong 1999–2000
Jaggi & Tsui 2007	BOD characteristics	discretionary accrual	Hong Kong 1995–99
Jeong & Rho 2004	auditor size	discretionary accrual	Korea 1994–98
Kao & Chen 2004	BOD size & characteristics	discretionary accrual	Taiwan; year not reported
Klein 2002	audit committee and BOD characteristics	abnormal accrual	US 1992–93
Krishnan 2003a	auditor specialization	abnormal accrual	US 1989–98
Lee <i>et al.</i> 2006	corp governance	discretionary accrual	US 1991–2004
Li & Lin 2005	non-audit fee	restatement	US 2000
Lin <i>et al.</i> 2006	audit committee characteristics	restatement	US 2000
Maijor & Vanstraelen 2006	auditor size	discretionary accrual	France, Germany, UK 1992–2000
Menon & Williams 2004	audit committee characteristics	abnormal accrual	US 1998–99
Mitra 2007	auditor fees	discretionary accrual	US 2000
Myers <i>et al.</i> 2003	auditor tenure	abnormal accrual	US 1988–2000
Park & Shin 2004	board characteristics	abnormal accrual	Canada 1996–97
Peasnell <i>et al.</i> 2005	audit committee & board characteristics	abnormal accrual	UK 1993–96
Piot & Janin 2007	audit committee & auditor characteristics	discretionary accrual	France 1999–2001
Raghunandan <i>et al.</i> 2003	non-audit fee	restatement	US 2001
Rahman & Ali 2006	BOD & audit committee char and auditor size	discretionary accrual	Malaysia 2002–03
Reitenga & Tearney 2003	BOD & audit committee characteristics	discretionary accrual	US 1987–96
Reynolds <i>et al.</i> 2004	auditor fees & size	discretionary accrual	US 2000
Van der Zahn & Tower 2004	audit committee characteristics	abnormal accrual	Singapore 2000–01
Yang & Krishnan 2005	audit committee characteristics	abnormal accrual	US 1996–2000

Note: Studies included must meet all of the following criteria:

1. The dependent variable must be measures based on abnormal (discretionary) accrual, financial statement restatement or reporting fraud.
2. The independent variables in a multivariate model must include measures of audit quality and effectiveness of corporate governance relating to the board of directors and audit committee.
3. The test statistics or *p*-value for computing the effect size must be reported.

large number of studies are excluded from the meta-analysis because they do not meet the criteria specified above. Table 2 lists these excluded studies and states the reason for their exclusion.

Following prior meta-analysis studies in accounting (e.g., Hay *et al.*, 2006; Kinney & Martin, 1994), we use the Stouffer combined test to summarize the effects on earnings management of various independent variables, which are reported with a *t*-statistic, χ^2 -statistic, or *p*-value in individual existing studies. We convert all *t*-statistics and χ^2 -statistics to their corresponding *p*-values and then to Z-statistics as the measure of effect size. The individual Z-statistics are then combined using the following formula (Wolf, 1986: 20):

$$\text{Unweighted } Z_c = \frac{\sum Z}{\sqrt{N}}, \quad (2)$$

where *N* is the number of studies under review.

It may be argued that not all studies in a meta-analysis should be given equal weight. Some studies may use a small sample, while others may be based on a much larger sample. In the unweighted case, as is the case in Formula (2) above, studies with small samples could exert a much stronger effect on the results than warranted. Wolf (1986) recommends that both the unweighted and weighted *Z_c* be calculated. Therefore, the Stouffer combined test based on the sample-size weighted *Z_c* giving more weight to large samples is calculated as follows (Wolf, 1986: 40):

$$\text{Weighted } Z_c = \frac{\sum df * Z}{\sqrt{\sum df^2}}, \quad (3)$$

where *df* is the degrees of freedom associated with the statistic of each study.

Finally, there is a potential problem when including only published studies. While the manuscript review process helps ensure the quality of published studies, a publication bias may result from the tendency that studies with significant results or larger effect sizes are more likely to be published than those without significant results or with smaller effect sizes. This problem is commonly referred to as the 'file drawer' problem in that these unpublished studies are buried away in 'file drawers' (Hay *et al.*, 2006; Wolf, 1986).

To deal with publication bias, in a meta-analysis, the fail-safe number, *N_{fs}*, is calculated to show the number of studies failing to report significant results that would be needed to reverse a

conclusion about a significant relationship between the dependent and independent variables. Using the results of the Stouffer combined test, the fail-safe number is computed as follows (Rosenthal, 1991: 261):

$$N_{fs} = \left(\frac{k \times (k \times z^2 - 2.706)}{2.706} \right), \quad (4)$$

where *k* is the number of studies in the meta-analysis and *z* is the combined standard z-value for the meta-analysis. The robustness of a significant relationship as represented by its fail-safe number can be compared with a critical number of studies that may be filed away. Rosenthal (1991: 262) provides the following equation for calculating the critical number of studies:

$$\text{Critical number of studies} = (5 \times k) + 10, \quad (5)$$

where *k* is the number of studies in the meta-analysis. According to Clark-Carter (1997) and Rosenthal (1991), the file drawer issue is only a problem if the fail-safe number is not greater than the critical number of studies. Moreover, the fail-safe number and the critical number of studies only need be calculated for significant relationships.

RESULTS

Table 3 reports the main results of the meta-analysis of the effects of corporate governance and audit quality attributes on earnings management. For each attribute, we discuss its nature and hypothesized effect on earnings management (earnings quality or lack thereof), and the results from our meta-analysis.

Corporate governance

The role of corporate governance structure of a corporation in financial reporting is to ensure compliance with GAAP and to maintain the credibility of corporate financial statements. Common measures of corporate governance effectiveness that are the focus of prior research are related to the composition, expertise, and activity of the board of directors (BOD) and its audit committee (AC).

BOD independence

Fama & Jensen (1983) recognize the board of directors as the most important management

Table 2: Studies excluded

<i>Study</i>	<i>Different independent variable</i>	<i>Different dependent variable</i>	<i>No applicable data</i>
Abbott <i>et al.</i> 2006 Aboody <i>et al.</i> 2005 Adjaoud <i>et al.</i> 2007	discretionary accruals insider trading	audit fees accounting rate of return & market-based performance	
Aier <i>et al.</i> 2005 Akhigbe <i>et al.</i> 2005 Arnold <i>et al.</i> 2001 Arthaud-Day <i>et al.</i> 2006	CFO characteristics turnover of CEO, CFO, BOD and audit committee members	abnormal stock return additional audit work	
Ascioglu <i>et al.</i> 2005 Ashbaugh-Skaife <i>et al.</i> 2006 Ball & Shivakumar 2005 Ball & Shivakumar 2006	firm characteristics cash flows & stock returns audit opinions	market liquidity credit ratings	
Bartov <i>et al.</i> 2001 Beasley <i>et al.</i> 2000 Beatty & Weber 2006 Beekes <i>et al.</i> 2004 Blouin <i>et al.</i> 2007 Braiotta & Zhou 2006		goodwill writeoff reporting conservatism selection of new auditors audit committee alignment/change earnings surprise management	simple <i>t</i> -tests
Brown & Higgins 2001			
Butler <i>et al.</i> 2004 Carcello & Neal 2003	auditor opinions	management discussion of financial distress	
Charitou <i>et al.</i> 2007 Chen <i>et al.</i> 2001 Chia <i>et al.</i> 2007	auditor opinions auditor opinions		auditor size in different models
Chin <i>et al.</i> 2006 Chung & Kallapur 2003 Cohen <i>et al.</i> 2007 Cohen <i>et al.</i> 2004 Davidson & Neu 1993 Davidson <i>et al.</i> 2006 DeFond 1992 DeFond & Jiambalvo 1991 DeFond & Park 2001 DeZoort <i>et al.</i> 2003 El Mir & Seboui 2006 Ettredge <i>et al.</i> 1988 Fairfield <i>et al.</i> 2003 Fields & Keys 2003 Filatotchev <i>et al.</i> 2005 Francis 2006 Francis <i>et al.</i> 2004 Gaver & Paterson 2001 Geiger <i>et al.</i> 2005	client importance auditor changes abnormal accruals earnings attributes hiring of former auditors	earnings forecast earnings forecast error auditor changes accounting errors loss adjustment	literature review literature review
Givoly <i>et al.</i> 2007 Glaum <i>et al.</i> 2004 Goodwin & Seow 2002 Gul <i>et al.</i> 2003	firm characteristics	market value stock return accrual growth financial performance reporting conservatism stock market reaction	behavior experiment literature review literature review behavior experiment

Table 2: Continued

<i>Study</i>	<i>Different independent variable</i>	<i>Different dependent variable</i>	<i>No applicable data</i>
Gul <i>et al.</i> 2006 Habib 2006 Hartnett 2006 Haw <i>et al.</i> 2005 Healy & Wahlen 1999 Heninger 2001 Higgs & Skantz 2006 Hodge 2003 Hui & Fatt 2007 Iturriaga & Hoffmann 2005 Jenkins <i>et al.</i> 2006	auditor litigation	stock return earnings forecast stock return stock return earnings quality	literature review literature review literature review
Johl <i>et al.</i> 2007 Kanagaretnam <i>et al.</i> 2007 Karamanou & Vafeas 2005 Keasey & McGuinness 1991 Kim <i>et al.</i> 2003 Knechel & Payne 2001 Knechel & Vanstraelen 2007 Koh 2003	stock ownership interactive effect of event and auditor specialization auditor conservatism institutional ownership	auditor opinions info asymmetry earnings forecast earnings forecast audit report lag auditor opinions	 main auditor specialization effect not reported
Krishnan 2003b Kwak 2002 Lapointe-Antunes <i>et al.</i> 2006 Larcker & Richardson 2004	voluntary disclosure	stock market reaction	literature review
Larcker <i>et al.</i> 2007			three subsample clusterwise regressions; no control variables IVs are factor loading scores
Lee & Mande 2003	private securities litigation reform act		
Lee <i>et al.</i> 2003 Leng 2004 Leuz <i>et al.</i> 2003 Marnet 2007 Matsumoto 2002 McNichols 2000 Menon & Williams 1994	investor protection firm characteristics	auditor choice return on equity	 literature review literature review
Mitra & Cready 2005	institutional ownership	audit committee characteristics	
Nelson <i>et al.</i> 2003 Palmrose & Scholz 2004 Peasnell <i>et al.</i> 2000	restatement different accrual models		survey
Pergola 2005 Petra 2007 Phillips <i>et al.</i> 2003 Pincus & Rajgopal 2002 Rezaee <i>et al.</i> 2003 Rowland 2002 Ruddock <i>et al.</i> 2006 Saleh & Ahmed 2005	firm char. debt renegotiation	stock market reaction deferred tax expense reporting conservatism	literature review literature review literature review

Table 2: Continued

<i>Study</i>	<i>Different independent variable</i>	<i>Different dependent variable</i>	<i>No applicable data</i>
Sánchez-Ballesta & García-Meca 2005		audit opinions	
Sánchez-Ballesta & García-Meca 2007	stock ownership structure		
Schipper 1989			literature review
Shaw 2003		corporate disclosure ratings	
Shen & Chih 2005	firm characteristics		<i>t</i> or <i>p</i> -values not reported
Srinidhi & Gul 2007			
Srinivasan 2005	restatement		literature review
Staubus 2005			
Summers & Sweeney 1998	insider trading		
Teoh & Wong 1993		stock market reaction	
Vafeas 2005		small earnings increase & unexpected earnings	
Van Caneghem 2004		earnings rounding-up behavior	
Van der Zahn & Tower 2006		auditor fees	
Wang 2006	founding family ownership		
Wells 2002	CEO changes		
Wild 1996		stock market reaction	
Wright <i>et al.</i> 2006	firm characteristics		
Xie 2001		stock market reaction	
Yee 2006			analytical study
Yeo <i>et al.</i> 2002	management stock ownership		
Zhao & Millet-Reyes 2007		stock return	

Note: Studies are excluded if the independent variable is not defined as some measure of board or audit committee characteristics or audit quality, or the dependent variable is not based on abnormal accrual, financial statement restatement or fraud as the measure of earnings management (quality), with the variables actually used noted in the applicable cells. Studies are also excluded when the test statistics, *p*-value or similar data needed to compute the effect size for meta-analysis is not reported.

control mechanism. From an agency perspective, the ability of the board to function as an effective oversight of management in the areas of financial reporting rests upon its independence from management (Beasley, 1996). Therefore, it is assumed that effective governance and financial reporting quality increase with board independence (as measured by the proportion of outside or independent directors on the board). A slight majority of prior studies report a significantly negative relationship between earnings management and increased BOD independence (e.g., Beasley, 1996; Klein, 2002). However, Park & Shin (2004) and Peasnell *et al.* (2005) do not find a significant relationship. The meta-analytical results reported in Table 3 show that independence of the board has a significant negative relationship (at the 1% level) with

occurrence of earnings management, based on either unweighted or weighted Stouffer test. Also, the fail-safe number greatly exceeds the critical number of studies (2,285 versus 100), hence strongly supporting the hypothesis that earnings management decreases as the board independence increases as suggested by Fama & Jensen (1983).

BOD expertise

While a more independent board may intend to restrain earnings management, only outside or independent directors on the board with proper background may be able to do so. A director with financial expertise may have greater familiarity with how earnings can be managed and take necessary measures to curb earnings management. Relatively few existing studies examine this issue.

Table 3: Effect of audit quality and corporate governance variables on earnings management

Variable	Stouffer test using unweighted Z	Stouffer test using weighted Z	Number of studies	Fail- safe number	Critical number for drawer
Board of Directors (BOD) Independence	-4.904***	-4.386***	18	2285	100
BOD Expertise	-3.655***	-3.511***	3	38	25
BOD Stock Ownership	1.169	0.666	12	N/A	N/A
BOD Independent Chair	1.151	-0.271	10	N/A	N/A
Existence of Audit Committee (AC)	-1.254	1.037	6	N/A	N/A
AC Independence	-4.373***	-3.820***	14	1043	80
AC Number of Meetings	-3.788***	-2.487***	10	218	60
AC Size	-2.742***	-2.265**	7	85	45
AC Expertise	-3.812***	-2.444***	9	169	55
AC Stock Ownership	2.940***	2.983***	4	48	30
Auditor Tenure	-3.264***	-3.095***	7	166	45
Auditor Size	-3.567***	-5.501***	23	5892	125
Auditor Specialization	-5.438***	-4.901***	3	77	25
Auditor Independence					
Fee ratio	6.137***	5.423***	10	1076	60
Total fee	4.330***	3.101***	7	167	45
Audit fee	1.936**	0.232	4	N/A	N/A
Nonaudit fee	2.542***	1.557	7	N/A	N/A

Note: ** = significant at the 5% level, *** = significant at the 1% level.

One of the studies (Park & Shin, 2004) reports a significantly negative association between increased board financial expertise and earnings management, but another study (Xie *et al.*, 2003) finds a negative but insignificant relationship. Our meta-analysis results suggest that the board financial expertise is negatively related to earnings management (significant at the 1% level) using either the unweighted or weighted Stouffer combined test. The fail-safe number exceeds the critical number of studies (38 versus 25), supporting the negative relationship.

BOD stock ownership

A clear theoretical prediction about the effect of stock ownership by directors on the effectiveness of the board in monitoring management does not exist. Gul *et al.* (2002: 30) argue that 'managers of firms with low director ownership are expected to respond to accounting-based contracts by exploiting the latitude available in accounting procedures to either alleviate constraints or capitalize on available incentives suggesting a higher level of earnings management'. In other words, higher stock ownership by directors will reduce the occurrence of earnings management. However, a direct financial interest, such as stock

ownership by outside directors, may weaken the independence of directors and their effectiveness in monitoring management decisions, including in the area of financial reporting. Prior studies provide mixed evidence on the effect on earnings management of directors' stock ownership in the firm. For example, Gul *et al.* (2002) document a significantly negative association between directors' stock ownership and earnings management. In contrast, Peasnell *et al.* (2005) report a positive, though not significant, association. Our meta-analysis suggests no significant relationship exists between stock ownership by directors and earnings management. Further research is probably warranted.

BOD independent chair

Another important characteristic of the board is whether the position of the Chief Executive Officer (CEO) is separate from the position of the chairperson of the board. One of the important roles played by the chairperson of the board is to run the board meetings and oversee the process of hiring, evaluating, firing and compensating the CEO. Jensen (1993) argues that it creates a conflict of interest for the CEO to serve as the board chair and perform the oversight function related to this

process. He argues that it is important to separate the CEO and the chairperson positions for the board to provide effective monitoring. Therefore, it is expected that the separation of the positions of CEO and board chairperson reduces earnings management. However, none of the existing studies report any significant relationship. The meta-analysis results presented in Table 3 also do not indicate any significant relationship, consistent with prior research.

AC existence

In order to more efficiently perform their duties, the board of directors often delegates the responsibility for overseeing financial reporting to an audit committee. The audit committee is viewed as enhancing the board of directors' capacity to monitor management in the financial reporting process by providing more detailed knowledge and understanding of financial statements and other financial disclosures issued by the company. The existence of an audit committee may be perceived as indicating higher quality monitoring and should reduce the occurrence of opportunistic earnings management. Empirical studies to date reported mixed results. For example, while Bédard *et al.* (2004) and Jaggi & Leung (2007) report a significantly negative relationship between earnings management and the existence of an audit committee, all the other existing studies either fail to find a significant relationship or find a significant but positive (contrary to expectation) relationship. Our meta-analysis shows that there is no significant relationship between the existence of an audit committee and earnings management based on either unweighted or weighted Stouffer combined tests. Further research may help clarify the issue.

AC independence

The effect of independence of audit committee members has been examined in most of the prior studies on earnings management. A common expectation is that a more independent audit committee would provide more effective oversight of the financial reporting process and ensure better quality of earnings reported by the firm by restraining opportunistic earnings management (BRC, 1999; SEC, 1999). However, while such expectation is easily understandable, the positive effect of audit committee independence on

financial reporting quality is not consistently supported in prior studies. For example, while Klein (2002) and Abbott *et al.* (2004) document that the level of audit committee independence is negatively associated with earnings management, Lin *et al.* (2006), Reitenga & Tearney (2003), and Xie *et al.* (2003) do not find such a significant relationship. The meta-analysis results reported in Table 3 show a highly significant negative relationship (at the 1% level) between AC independence and earnings management, consistent with the expected effect. Furthermore, the fail-safe number greatly exceeds the critical number of studies (1,043 versus 80), providing strong support for the positive effect of an independent AC on financial reporting quality.

AC meetings

An important objective for an audit committee is to provide its members with sufficient time to perform their duties of monitoring their firm's financial reporting process. While it is not mandated by the SEC, the BRC (1999) recommends that audit committees meet at least once quarterly and discuss financial reporting quality with the external auditor. The number of meetings (a proxy for diligence) is used in prior research because inactive audit committees are unlikely to monitor management effectively (Menon & Williams, 1994). The prior research provides inconsistent evidence on the issue. For example, Lin *et al.* (2006) and Xie *et al.* (2003) report a negative association between earnings management and the number of AC meetings. In contrast, Bédard *et al.* (2004), and Yang & Krishnan (2005) fail to find such an association. Our meta-analysis, as reported in Table 3, shows a significant negative relationship (at the 1% level) between earnings management and the number of AC meetings, based on either unweighted or weighted tests. The fail-safe number exceeds the critical number of studies by a wide margin (218 versus 60), supporting a strong positive effect of an active audit committee in ensuring financial reporting quality.

AC size

Encouraged by the BRC (1999), the SEC (1999) mandates that audit committees consist of a minimum of four directors. A larger audit committee represents greater resources and talents to rely on in overseeing the financial reporting

process. Empirical studies provide mixed evidence on the impact of audit committee size on earnings management. Xie *et al.* (2003) find no significant association between the number of directors on the audit committee and earnings management. Similarly, Abbott *et al.* (2004) and Lin *et al.* (2006) find no impact of audit committee size on earnings restatement. On the other hand, Yang & Krishnan (2005) find that audit committee size is negatively associated with earnings management. The results of the meta-analysis presented in Table 3 show a significant negative association (at the 1% level for the unweighted and 5% level for the weighted test) between audit committee size and earnings management. The fail-safe number also substantially exceeds the critical number of studies (85 versus 45), supporting the importance of having large audit committees.

AC expertise

The SEC (1999) requires that every audit committee includes at least one member qualified as 'financial expert' and that all committee members must be financially literate. As required by the Sarbanes-Oxley Act, the SEC adopted in 2003 a final definition for audit committee financial experts and it generally includes knowledge and experience in financial accounting and reporting, auditing, and similar functions (see <http://www.sec.gov/rules/final/33-8177.htm> for details on the attributes of so-called financial experts). DeZoort & Salterio (2001) argue that the audit committee's financial expertise (specifically auditing knowledge) increases the likelihood that detected material misstatements will be communicated to the audit committee and corrected in a timely fashion. However, aside from the SEC's definition, there is no agreement on what constitutes 'financial expertise' or on how to measure it. The empirical evidence is mixed. Abbott *et al.* (2004) and Bédard *et al.* (2004), among others, report a negative association between the audit committee's financial expertise and occurrence of earnings management. However, many other studies do not find such a significant relationship (e.g., Lin *et al.*, 2006). The results in Table 3 suggest that, consistent with expectation, the relationship between earnings management and AC expertise is significantly negative at the 1% level based on either unweighted or weighted tests. The fail-safe number greatly exceeds the critical number of studies (169

versus 55). Therefore, the evidence is strong on the significant effect of financial expertise.

AC stock ownership

Some (e.g., Gul *et al.*, 2002) suggest that stock ownership by directors (not necessarily on the audit committee) would align their interests with stockholders' and provide incentives to monitor management. However, stock ownership by AC members may weaken their independence and, thus, reduce the effectiveness of the audit committee in monitoring management in the financial reporting process. As a result, such stock ownership may actually increase the occurrence of earnings management, *ceteris paribus*. Empirical evidence is mixed. Choi *et al.* (2004) and Yang & Krishnan (2005) document that the extent of stock ownership by audit committee members increases the likelihood of earnings management. In contrast, Lin *et al.* (2006) do not find a significant relationship between shares owned by AC members and occurrence of earnings management. The meta-analysis results in Table 3 indicate that share ownership by AC members has a significant positive relationship (at the 1% level based on either unweighted or weighted tests) with earnings management. The results suggest the detrimental effect of stock ownership by AC members in providing effective monitoring of management in the financial reporting process. Also, the fail-safe number exceeds the critical number of studies (48 versus 30). The evidence is in stark contrast to the non-significant effect of stock ownership by the full board on earnings management as discussed above.

Audit quality

The role of auditing in ensuring the quality of reported earnings has come under considerable scrutiny due to recent corporate accounting scandals. 'Audit quality differences result in variation in credibility offered by the auditors, and in the earnings quality of their audit clients. Because auditor quality is multidimensional and inherently unobservable, no single auditor characteristic can be used to proxy for it' (Balsam *et al.*, 2003: 71). In this meta-analysis, we review the relationships between earnings management and several attributes of audit quality commonly investigated in prior studies.

Auditor tenure

Prior research suggests that auditor independence decreases as the length of auditor tenure increases (Beck *et al.*, 1988; Lys & Watts, 1994). The impaired independence results in poor audit quality and allows for greater earnings management (resulting in lower earnings quality). On the other hand, others claim that as auditor tenure increases, the auditor is better at assessing risk of material misstatements by gaining experience and better insights into the client's operations and business strategies as well internal controls over financial reporting (e.g., Arens *et al.*, 2005). Of the existing studies reviewed in this paper, only two (Myers *et al.*, 2003; Yang & Krishnan, 2005) report a significant negative relationship between auditor tenure and earnings management. The remaining studies report the relationship to be negative but not significant. The meta-analysis results in Table 3 indicate that auditor tenure has a significant negative relationship (at the 1% level) with earnings management using either unweighted or weighted tests. Also, the fail-safe number greatly exceeds the critical number of studies (166 versus 45). Hence, there is strong evidence to suggest that as the auditor's tenure increases, the benefit of the greater experience and better insight into the client's operations and business strategies seem to outweigh the potential independence impairment.

Auditor size

A number of studies examine whether auditor brand name, measured by auditor size (Big 6/5/4), is associated with earnings quality. For example, Becker *et al.* (1998) and Francis *et al.* (1999) argue that Big 6 auditors are better able to detect earnings management because of their superior knowledge, and act to curb earnings management to protect their reputation. Also, Krishnan (2003a) argues that, besides more resources and expertise to detect earnings management, the large audit firms also have greater incentives to protect their reputation due to their larger client base. However, empirical evidence on the issue is mixed. For example, while Francis *et al.* (1999) and Becker *et al.* (1998) report that the use of Big 6 auditors is associated with less earnings management, others (e.g., Antle *et al.*, 2006; Lin *et al.*, 2006) find evidence to the contrary. The meta-analysis results presented in Table 3 show a significant negative relationship (at the 1% level) between the use of Big 6/5/4 auditors and earnings

management, consistent with expectation. The evidence is very strong as suggested by the exceptionally large fail-safe number of 5,892, compared with the critical number of 125.

Auditor specialization

In addition to auditor brand name, some recent studies (e.g., Balsam *et al.*, 2003) argue that an industry specialist auditor offers a higher level of assurance than does a non-specialist because of the specialist auditor's knowledge of the industry and its accounting. Therefore, the use of an auditor with industry specialization will help curb earnings management. Three existing studies examine this relationship. Balsam *et al.* (2003) and Krishnan (2003a) report a negative association, but Chen *et al.* (2005) find a positive relationship. The meta-analysis results presented in Table 3 also show a significant negative relationship (at the 1% level using either unweighted or weighted tests) between earnings management and use of industry specialist auditor. So, empirical evidence to date suggests the positive benefit of using a specialist auditor in improving earnings quality. The conclusion is supported by the wide margin of the fail-safe number over the critical number of studies (77 versus 25).

Auditor independence

Prior studies contend that high fees paid by a company to its external auditor increase the economic bond between the auditor and the client and thus the fees may impair the auditor's independence (e.g., Frankel *et al.*, 2002; Li & Lin, 2005). The impaired independence results in poor audit quality and allows for greater earnings management (resulting in lower earnings quality). However, there is no agreement on how to measure this economic bond. Prior studies have used a number of variables: fee ratio (non-audit fee over total fee), total fees, and separate audit and non-audit fees.

When fee ratio is used, all prior studies reviewed report a positive association, although some (e.g., Huang *et al.*, 2007; Raghunandan *et al.*, 2003) find the relationship non-significant. The results reported in Table 3 indicate a significant positive relationship (at the 1% level) between fee ratio and occurrence of earnings management. Also, the fail-safe number greatly exceeds the critical number of studies (1,076 versus 60). The results are similar

Table 4: Subgroup analysis of selected variables

<i>Variable</i>	<i>Stouffer test using unweighted Z</i>	<i>Stouffer test using weighted Z</i>	<i>Number of studies</i>	<i>Fail-safe number</i>	<i>Critical number for drawer</i>
Board of Directors (BOD) Independence					
US	-2.848***	-1.892**	6	42	40
World	-3.992***	-3.684***	12	710	70
BOD Stock Ownership					
US	1.016	0.18	5	N/A	N/A
World	0.672	1.706**	7	N/A	N/A
Audit Committee Independence					
US	-3.987***	-2.329***	9	153	55
World	-1.968**	-2.194**	5	39	35
Auditor Size					
US	-5.065***	-5.546***	9	911	55
World	-0.51	-0.414	14	N/A	N/A
Level of Audit Committee Independence					
Complete Independence	-3.352***	-1.871**	6	41	40
Proportional Independence	-2.764***	-2.447***	9	170	55

Note: ** = significant at the 5% level, *** = significant at the 1% level.

when total fee is used. The fail-safe number also far exceeds the critical number of studies (167 versus 45). Thus, the evidence strongly supports the negative effect on earnings quality (or lack thereof) of a strong economic bond between the client and the external auditor, as measured by high total fees paid to the auditor or the high fees paid for non-audit services, relative to total fees.

Most prior studies also use separate fees for audit and non-audit services, usually in the same model. The results on the relationship between audit fee and earnings management are mixed. Some prior studies report a negative relationship (e.g., Frankel *et al.*, 2002) but Antle *et al.* (2006) and Lin *et al.* (2006) find the relationship to be positive. The results in Table 3 show a significant positive relationship when using the unweighted Stouffer test but a non-significant relationship when using the weighted test, reflecting the mixed results in prior studies. The results seems to be consistent with the notion that when the auditor provides a better quality audit, as reflected in a higher audit fee, earnings management is less likely. However, the results would not be consistent with the argument that higher fees, regardless of whether they are for audit or non-audit services, increase the economic bond between the auditor and the client, which in turn will result in poor audit quality and more earnings management.

As for non-audit fees, with the exception of Antle *et al.* (2006), who show a significant

negative relationship, all the other studies report a positive relationship but some of the studies find the relationship non-significant. The results in Table 3 again show a significant positive relationship when using the unweighted Stouffer test but a non-significant relationship when using the weighted test, reflecting the mixed results in prior studies. Overall, our results suggest that non-audit fees per se are less important than their relationship to the total fees paid to the auditor in determining earnings management.

Another benefit of meta-analysis is to perform subgroup analysis to determine the effect of potential moderators on the relationship between an independent variable and the dependent variable. As an illustration, Table 4 shows the results on four selected variables based on the US versus the world and the different levels of AC independence (complete independence versus proportional independence). These variables were chosen because they have a sufficient number of studies to perform such a subgroup analysis for each potential moderator. While the independence of the board of directors is significant overall, the effect is more profound in countries other than the US. The opposite can be observed about the independence of the AC: the effect is more pronounced in the US than in other countries. Similarly, the effect of auditor size as a deterrent to earnings management is more pronounced in the US than in other countries. In fact, the effect is not

significant when examined in other countries. Also, the share ownership of the board of directors has no significant effect on earnings management, in either the US or other countries. Finally, both levels of audit committee independence have a significant effect in reducing earnings management.

CONCLUSIONS

Earnings management is of great concern to corporate stakeholders. Despite the popularity of the topic, empirical evidence on the effect of audit quality and corporate governance is rather inconsistent. Our literature searches uncovered a large number of studies on earnings management and subjected 48 studies to meta-analysis. The remaining studies were excluded because of the use of dependent variables other than measures of abnormal (discretionary) accrual, financial statement restatement or reporting fraud. Studies were also excluded if the independent variables did not include measures of corporate governance or audit quality, or requisite data for computing the effect size. Using the Stouffer combined test, this meta-analysis has identified consistent effects of a large number of audit quality and corporate governance variables. Given the relatively small number of studies published to date on these two important issues, ample opportunities exist for more research on the effect of corporate governance effectiveness and audit quality on earnings management.

The purpose of meta-analysis is to take stock and provide directions for future research rather than providing the final word (Wolf, 1986). Based on the results summarized in Table 3, the effects of the board of directors' stock ownership, existence of an audit committee, and the separation of the board chairperson position from the CEO position on earnings management are potential areas for future research. Another prospective research stream is to examine the moderating effect of certain variables such as country as discussed above.

Also, with the passage of the Sarbanes-Oxley Act (SOX) in the US and similar Acts in other countries, many of the variables tested in this meta-analysis have become mandatory. It would be interesting to see whether the results are different pre- and post-SOX. Currently, all but one (Huang *et al.*, 2007) study reviewed in this meta-analysis used data collected for periods before SOX. When more studies with post-SOX data become available,

testing the effects of time and other moderators will shed additional light on earnings management.

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